Mani Chakradhar Chanda

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PH: 660-528-1906 DataEngineer

Professional Summary

- Possessing 5+ years of comprehensive IT experience, including expertise in cloud computing (AWS,GCP) and on-premises clusters.
- Demonstrating proficient knowledge of both ETL and ELT data ingestion frameworks.
- Utilizing various data ingestion strategies, including Full Refresh, Incremental, and SCD Type2.
- Having exposure to end-to-end pipeline processes, starting from data ingestion to consumption.
- Experienced in both batch and real-time data ingestion using Spark and Sqoop.
- Participated in migration projects from on-premises to cloud environments and contributed to building features from the ground up.
- Skilled in translating business requirements into SQL queries and deriving meaningful insights.
- Proficient in ingesting data from RDBMS systems like MySQL, Postgres, and Oracle using Sqoop and Spark JDBC.
- Implemented data cleansing, data quality checks, and the application of business rules using Shell scripts and Spark.
- Leveraged Spark DataFrame transformations to map business rules and apply actions on top of transformations.
- Worked with sensitive data and implemented data encryption measures.
- Experienced in using BI reporting tools such as Tableau, Microsoft Excel, and PowerBI.
- Developed complex SQL queries for constructing KPI tables using aggregations, analytical functions, and window functions.
- Possesses a strong understanding of Hadoop YARN architecture and proficiency with tools like Sqoop, Hive, Spark, and Oozie.
- Knowledgeable in Git flow and branching strategies.
- Familiar with Git Ops (CI/CD) pipelines and has exposure to writing and executing CI/CD pipelines.
- Well-versed in the software development life cycle (SDLC).
- Experienced with Agile processes and has worked in both Scrum and Kanban methodologies.

Tools & Frameworks Professional Experience

- On-Prem : Hadoop (CDH), Hive, YARN, HDFS, MapReduce, Sqoop, Oozie, Spark
- GCP : GCS, Cloud Composer, Dataproc,BigQuery, Airflow
- Databases : MySQL, Postgres, Oracle, Microsoft SQL Server
- Data Warehouse : Hive, Redshift, Snowflake
- Messaging : Kafka
- No-SQL : HBase
- Scheduler : Oozie, Airflow
- Languages : Python, pandas, Java, core java, SQL, Shell script, JavaScript
- Version Control : Git, Bitbucket, Gitlab
- CI/CD : Jenkins, Gitlab
- ProjectManagement : Jira, Confluence
- Operating Systems : Linux

Professional Experience

Data Engineer

Infinite Computer Solutions **July 2023 to present** **Responsibilities:**

- Understanding requirements and formulating detailed designs for the modules.
- Collaborated closely with stakeholders to grasp project requirements and devised comprehensive designs for data engineering modules on the AWS cloud platform.
- Leveraged Python to develop robust serverless functions, catering to diverse data processing scenarios, including the dynamic creation of Directed Acyclic Graphs (DAGs), processing data stored in Amazon S3, and seamlessly loading data into Amazon Redshift.
- Designed and implemented intricate SQL queries and stored procedures to meet Change Data Capture (CDC) and Slowly Changing Dimension (SCD) requirements, establishing refined and accessible data layers within Amazon Redshift.
- Engineered custom functions and authorized views in Amazon Redshift to enhance data sharing capabilities across various AWS projects.
- Developed infrastructure provisioning scripts utilizing AWS SDKs and Terraform, ensuring efficient and scalable cloud resource management.
- Spearheaded the development of a dynamic DAG factory framework, enabling the generation of adaptive DAGs through AWS Lambda functions, DAG templates, and configuration files.

- Created Airflow DAGs within AWS Step Functions for orchestrating data pipelines, utilizing Bash, Python, Amazon Redshift, and Amazon EMR (Elastic MapReduce) operators.
- Established configuration files for the construction of Extract, Transform, Load (ETL) pipelines, ensuring seamless data flow and transformation.
- Authored Spark code for data ingestion from diverse sources like Oracle, PostgreSQL, catering to various data processing scenarios, including Full Refresh, Incremental Updates, and data curation use cases.
- Conducted in-depth data analysis on source data, shaping the target data model to meet specific business requirements.
- Thoroughly executed unit tests and integration tests to validate the reliability and accuracy of data pipelines.
- Designed and developed visually appealing dashboards using AWS QuickSight and Tableau, providing actionable insights to stakeholders.
- Maintained comprehensive project documentation in Confluence and ensured timely updates of Jira user stories.

Environment: PySpark, AWS Glue, AWS Lambda, Amazon Redshift, Amazon S3, AWS Step Functions, AWS EMR, Python, SQL, GitLab, Jira, Confluence.

Data Engineer

Skillo Inc **January 2022 to July 2023**

Responsibilities:

- Gaining an understanding of requirements and formulating detailed designs for the modules.
- Collaborated closely with stakeholders to decipher project requirements, leading to the creation of comprehensive module designs tailored to AWS cloud solutions.
- Developed efficient Python-based AWS Lambda functions, addressing a wide array of data processing scenarios, such as the dynamic generation of Directed Acyclic Graphs (DAGs) and eventdriven triggers.
- Designed and implemented AWS Glue ETL jobs, leveraging the power of Spark Core and SQL for intricate data processing tasks.
- Crafted optimized SQL queries within Amazon Redshift for data refinement, performing joins, aggregations, and constructing key performance indicator (KPI) tables.
- Engineered custom User-Defined Functions (UDFs) to enhance data transformations within Amazon Redshift.
- Automated infrastructure provisioning by scripting with AWS SDKs, ensuring scalability and efficiency.

- Orchestrated complex job workflows using Apache Airflow DAGs, seamlessly integrating AWS services within the cloud ecosystem.
- Spearheaded the migration of Hadoop components, including Hive, Sqoop, and Spark, to AWS, optimizing data processing capabilities.
- Created configuration files to streamline the development of Extract, Transform, Load (ETL) pipelines within AWS.
- Developed Spark code for versatile data ingestion strategies, encompassing Full Refresh, Slowly Changing Dimension Type 2 (SCD Type 2), and Incremental, tailored for the AWS environment.
- Conducted rigorous unit tests and integration tests to validate the reliability and accuracy of AWS data processes.
- Produced comprehensive documentation in Confluence and maintained up-to-date Jira stories for AWS-based projects, ensuring transparency and collaboration. **Environment**: PySpark, Shell Script, Python, SQL, Confluence.

Big Data Engineer

Genems Solutions **February 2020 to July 2021**

Responsibilities:

- Developing Sqoop Jobs for the ingestion of data from RDBMS systems such as Oracle.
- Creating various types of Hive tables, including Managed and External tables.
- Examining data and business use cases and constructing data models that involve partitioning, bucketing, and skewing.
- Crafting HQL files to construct aggregated tables, enrich data, and implement User-Defined Functions (UDFs).
- Developing PySpark applications to access and clean up data stored in HDFS.
- Creating Spark code to execute intricate transformations using Spark SQL and UDF functions.
- Submitting Spark applications in lower environments and conducting log analysis.
- Overseeing and analyzing issues in production environments.
- Loading data into Hbase using Spark.
- Designing Oozie workflows for the orchestration and scheduling of jobs.
- Creating data marts tailored to specific client needs for sharing with stakeholders.

Environment: Cloudera (CDH), Hadoop, Spark, Sqoop, Hive, HDFS, Oozie, Hbase.

Academic Details

- Master’s in Information Technology Management at Lindsey Wilson College (class of May 2023)
- Bachelor of Science(Computer Science) from Acharya Nagarjuna University ,Guntur(class of June 2020)